

Task 1.2 – SAP S/4HANA Module Mapping

Project Title

SAP S/4HANA Multi-Module ERP Implementation for Hospitality Operations: Integrating FI, CO, SD, and MM Modules for Operational and Financial Excellence – A Case Study of Resort Operations in Northern Japan

1. Purpose of This Document

The purpose of this document is to map the business processes and analytical systems developed in this project to the corresponding SAP S/4HANA modules.

This project aims to replicate core SAP ERP functionalities within a hospitality environment by integrating Financial Accounting (FI), Controlling (CO), Sales and Distribution (SD), and Materials Management (MM) modules. The project also includes an Analytics and Business Intelligence layer to support management decision-making.

The mapping demonstrates how the implemented solution reflects real SAP business processes and enterprise system architecture.

2. SAP Module Mapping Overview

SAP Module	SAP Functionality	Project Implementation
FI-AR	Accounts Receivable Management	AR Aging Analysis and Collection Monitoring
FI-GL	General Ledger Accounting	Revenue Recognition and Financial Tracking
CO-CCA	Cost Center Accounting	Cost Center Analysis and Budget Monitoring
CO-PA	Profitability Analysis	Segment Profitability Analysis
SD-SLS	Sales Processing	Sales Performance Analytics
SD-BIL	Billing Management	Revenue and Billing Tracking
MM-PUR	Procurement Management	Purchase Order Management
MM-IM	Inventory Management	Inventory Tracking and Stock Control

3. Financial Accounting (FI) Module Mapping

FI-AR (Accounts Receivable)

SAP Functionality:

- Customer invoice management
- Open item tracking
- Collection monitoring
- AR aging analysis

Project Implementation:

- AR Aging Report
- Outstanding Balance Monitoring
- Collection Rate Analysis
- Customer Payment Behavior Analysis

Simulated SAP Transactions:

- FBL5N – Customer Line Item Display
- FD10N – Customer Balance Display

Expected Outputs:

- Aging buckets (Current, 31-60, 61-90, 90+ Days)
- Collection KPIs
- Overdue Customer Monitoring

FI-GL (General Ledger)

SAP Functionality:

- Financial postings
- Revenue tracking
- Financial reporting

Project Implementation:

- Revenue Recognition System
- Monthly Revenue Reporting

- Financial Period Closing Reports
- Cash Flow Analysis

Simulated SAP Transactions:

- FB50 – G/L Posting
- FAGLB03 – G/L Account Balance Display

Expected Outputs:

- Revenue Trends
- Monthly Financial Reports
- Cash Flow Reports

4. Controlling (CO) Module Mapping

CO-CCA (Cost Center Accounting)

SAP Functionality:

- Cost center planning
- Budget monitoring
- Cost control

Project Implementation:

- Cost Center Structure Design
- Budget vs Actual Analysis
- Cost Monitoring Dashboard

Simulated SAP Transactions:

- KS01 – Create Cost Center
- KSB1 – Cost Center Report

Cost Centers:

- Accommodation Operations
- Food and Beverage
- Facilities and Maintenance
- Sales and Marketing
- Administration
- Finance and Accounting

Expected Outputs:

- Budget Variance Reports

- Cost Center Performance Reports
- Cost Allocation Analysis

CO-PA (Profitability Analysis)

SAP Functionality:

- Profitability reporting
- Segment performance analysis

Project Implementation:

- Profitability Analysis by Revenue Category
- Seasonal Profitability Analysis
- Customer Segment Profitability Analysis

Simulated SAP Transactions:

- KE30 – Profitability Report

Expected Outputs:

- Contribution Margin Analysis
- Profitability by Segment
- Seasonal Profitability Trends

5. Sales and Distribution (SD) Module Mapping

SD-SLS (Sales Processing)

SAP Functionality:

- Sales management
- Sales performance monitoring
- Customer analysis

Project Implementation:

- Sales Performance Tracking
- Customer Analytics
- Revenue by Channel Analysis

Simulated SAP Transactions:

- VA03 – Display Sales Order

Expected Outputs:

- Sales Trends
- Customer Segmentation
- Revenue by Category

SD-BIL (Billing)

SAP Functionality:

- Billing and invoice management
- Revenue posting

Project Implementation:

- Revenue Recognition
- Billing Analysis
- Sales Cycle Monitoring

Simulated SAP Transactions:

- VF03 – Display Billing Document

Expected Outputs:

- Billing Reports
- Revenue Tracking Reports
- Sales Cycle Analytics

6. Materials Management (MM) Module Mapping

MM-PUR (Procurement)

SAP Functionality:

- Purchase Order Management
- Vendor Management
- Procurement Tracking

Project Implementation:

- Vendor Master Data
- Purchase Order Tracking
- Procurement Analytics

Simulated SAP Transactions:

- ME21N – Create Purchase Order
- ME2N – Purchase Order Reporting

Expected Outputs:

- Purchase Spend Analysis
- Vendor Performance Reports
- Procurement KPIs

MM-IM (Inventory Management)

SAP Functionality:

- Inventory Tracking
- Stock Management
- Goods Movement Processing

Project Implementation:

- Inventory Monitoring System
- Stock Movement Tracking
- Reorder Point Management

Simulated SAP Transactions:

- MMBE – Stock Overview
- MIGO – Goods Movement

Expected Outputs:

- Inventory Status Dashboard
- Stock Alerts
- Inventory Optimization Reports

7. Analytics and Business Intelligence Layer

SAP Equivalent:

- SAP Analytics Cloud (SAC)
- SAP BusinessObjects BI

Project Implementation:

- Integrated Power BI Dashboard
- Executive KPI Dashboard
- Financial Dashboard
- Operational Dashboard

- Forecasting Dashboard

Integrated Data Sources:

- Sales Data
- Accounts Receivable Data
- Financial Data
- Cost Data
- Inventory Data

Expected Outputs:

- Executive Reporting
- KPI Monitoring
- Revenue Forecasting
- Cash Flow Forecasting
- Operational Insights

8. Cross-Module Integration

This project integrates multiple SAP modules to replicate real-world ERP business processes.

Procure-to-Pay Cycle (MM + FI)

Purchase Order → Goods Receipt → Vendor Invoice → Payment Processing

Order-to-Cash Cycle (SD + FI)

Sales Order → Billing → Customer Invoice → Payment Collection

Cost Management Cycle (MM + CO)

Material Consumption → Cost Allocation → Cost Center Reporting

Profitability Analysis Cycle (SD + CO)

Revenue Generation → Cost Assignment → Profitability Analysis

9. Conclusion

This project implements an integrated SAP S/4HANA-inspired ERP environment for the hospitality industry. The solution covers Financial Accounting (FI), Controlling (CO), Sales and Distribution (SD), and Materials Management (MM), while incorporating advanced analytics and forecasting capabilities.

By replicating key SAP business processes and transaction flows, the project demonstrates practical ERP implementation knowledge and provides a foundation for operational analysis, financial management, procurement control, and strategic decision-making.